# Leave-on hair preparations (leave-in conditioners, tonics/dressings, styling aids, hair sprays, color sprays): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

## Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for leave-on hair preparations (leave-in conditioners, tonics/dressings, styling aids, hair sprays, color sprays) in Cosmetics and Personal Care: Leave-on, evaluated on finished leave-on cosmetic, TOTAL CONTENT as placed on market (ug/kg of product as sold), on the technically-unavoidable-impurity basis every instrument in this family uses. Not a migration/leachate basis, not a dietary basis, not an ingredient purity specification.. 5 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

## 1 | How every number on this standard is set

HMTc Method v2.0, government-limit-only initial setting: every numeric cell is the strictest applicable sovereign maximum for this exact leave-on cosmetic form expressed as total content of the finished product as placed on market, or a disclosed read-across where the exact cell is unregulated, or a substantiated not-material determination, or a control. No cell is occurrence-derived, no cell is precautionary or feasibility-adjusted, and no cell is anchored on an ingredient or colour-additive purity specification.

## 2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 1,000 | government maximum level |
| Cadmium (Cd) | Tier 1 | 100 | government maximum level |
| Total arsenic (tAs) | Tier 2 | 500 | government maximum level |
| Inorganic arsenic (iAs) | Tier 1 | ≤ tAs | bounded by tAs (reflex speciation) |
| Mercury, total (tHg) | Tier 1 | 100 | government maximum level |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 10,000 | government maximum level |
| Tin (Sn) | Tier 2 | 3,750,000 | human-grade read-across (European Union) |
| Aluminium (Al) | Tier 2 | 560,000 | human-grade read-across (European Union) |
| Total chromium (Cr) | Tier 2 | 50 | human-grade read-across (AU) |
| Hexavalent chromium (Cr-VI) | Tier 2 | 5 | human-grade read-across (European Union) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 1,000 | government maximum level |
| Cadmium (Cd) | 100 | government maximum level |
| Total arsenic (tAs) | 500 | government maximum level |

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| | | |
|---|---|---|
| Inorganic arsenic (iAs) | ≤ tAs | iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Mercury, total (tHg) | 100 | government maximum level |
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 10,000 | government maximum level |
| Tin (Sn) | 3,750,000 | European Union — human-food analog |
| Aluminium (Al) | 560,000 | European Union — human-food analog |
| Total chromium (Cr) | 50 | AU — human-food analog |
| Hexavalent chromium (Cr-VI) | 5 | European Union — human-food analog |

## 4 | Cite this standard

Pendergrass, K. (2026). *Leave-on hair preparations (leave-in conditioners, tonics/dressings, styling aids, hair sprays, color sprays): HMTc standard*. Institute of Contaminant Standards.

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## References

[1] AU — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Sn, Al, Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

### STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's

own certification thresholds are not used as evidence for any factual claim made here.

## DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/leave-on-hair-preparations. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

## PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/leave-on-hair-preparations.

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